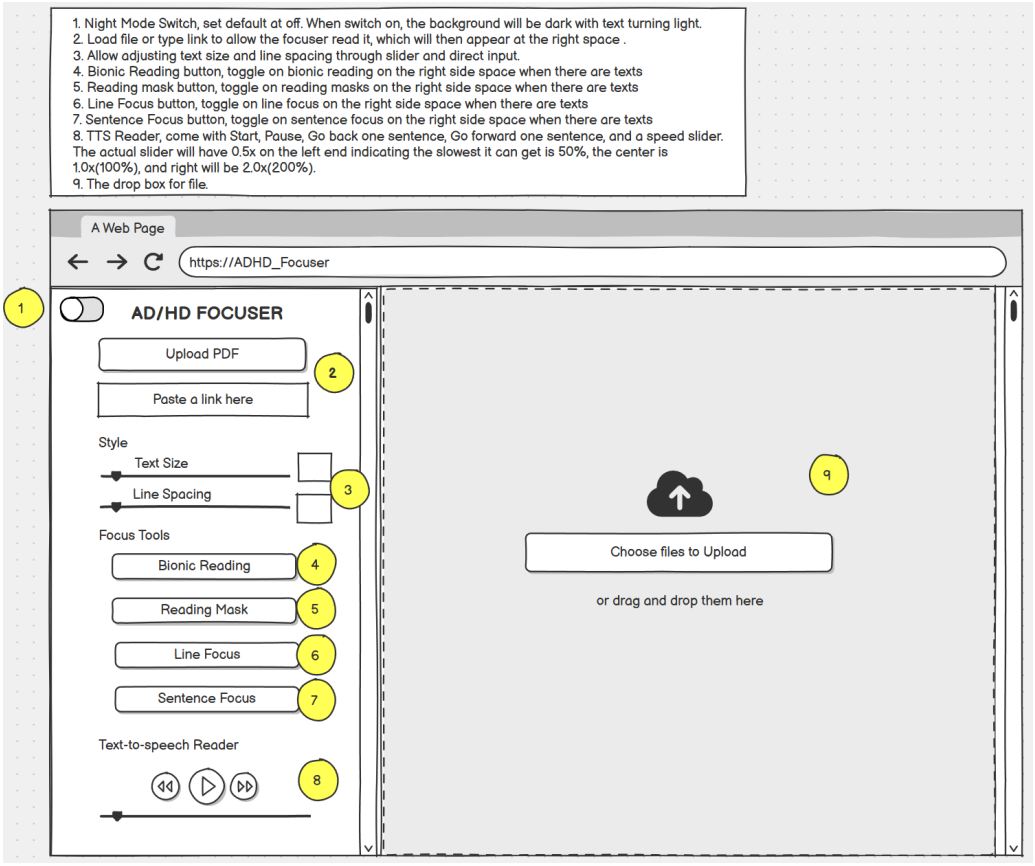
AD/HD
Project 1: Design

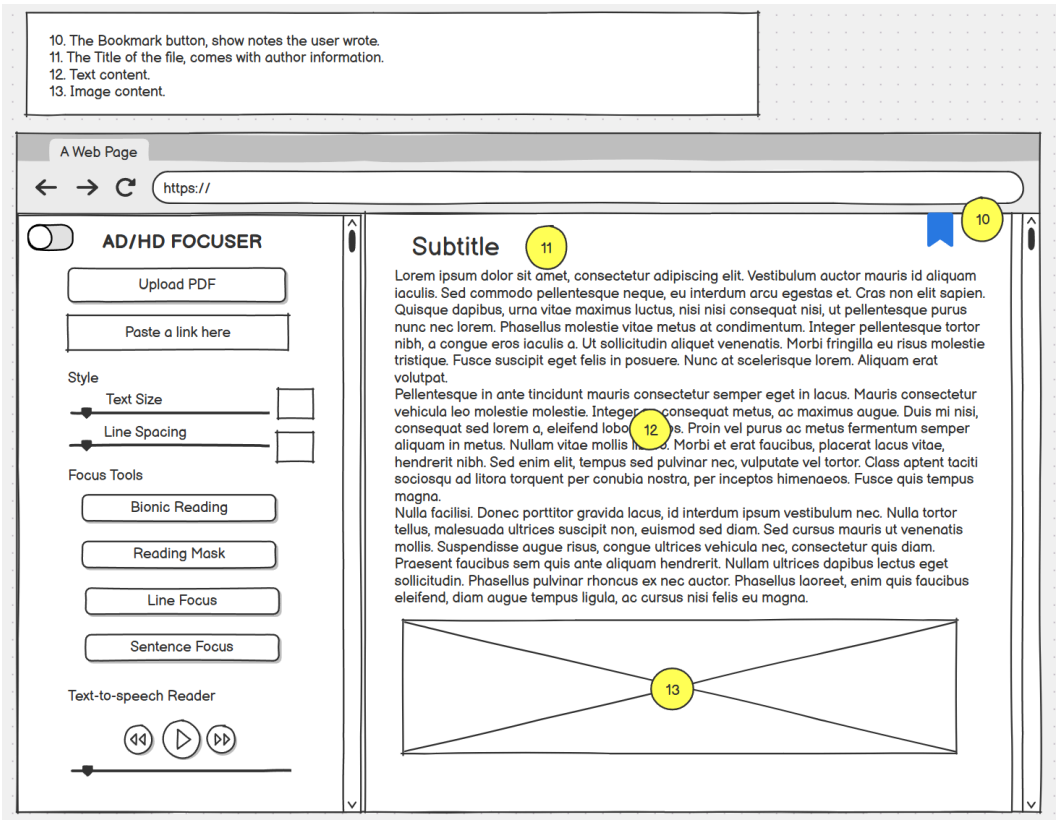
Yunxi Wang & Branden Secaira

UI Wireframes and Annotations

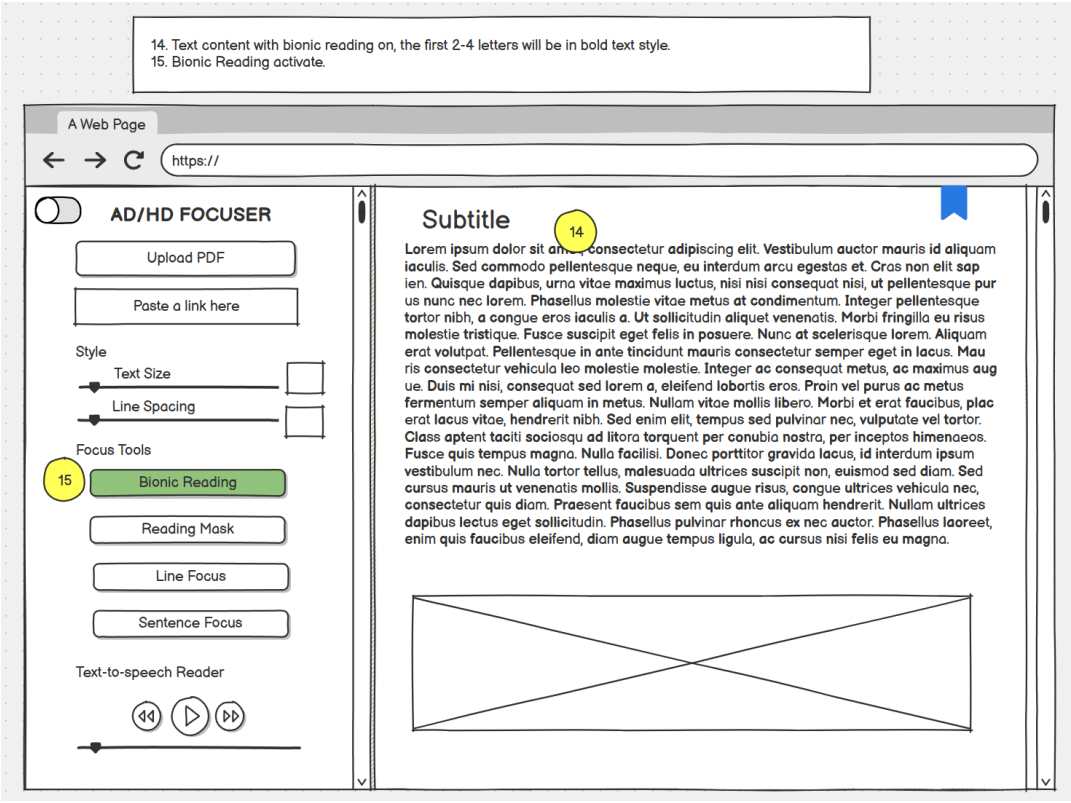
Default(No file) (Welkin)



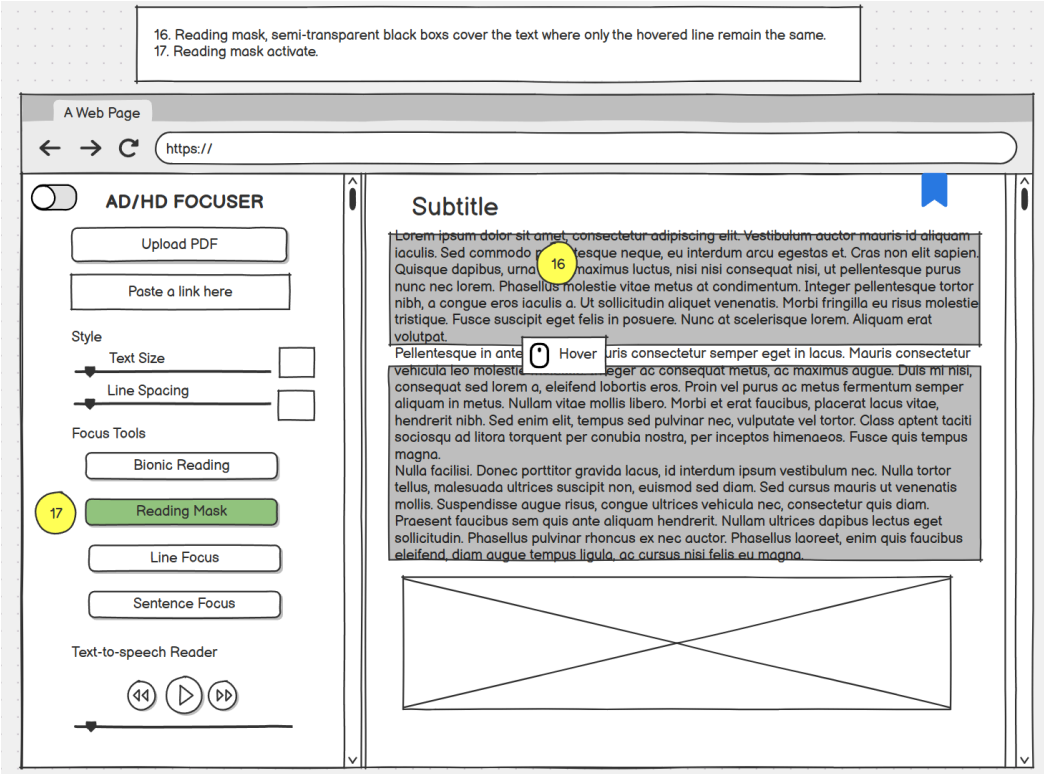
Default (With file) (Welkin)



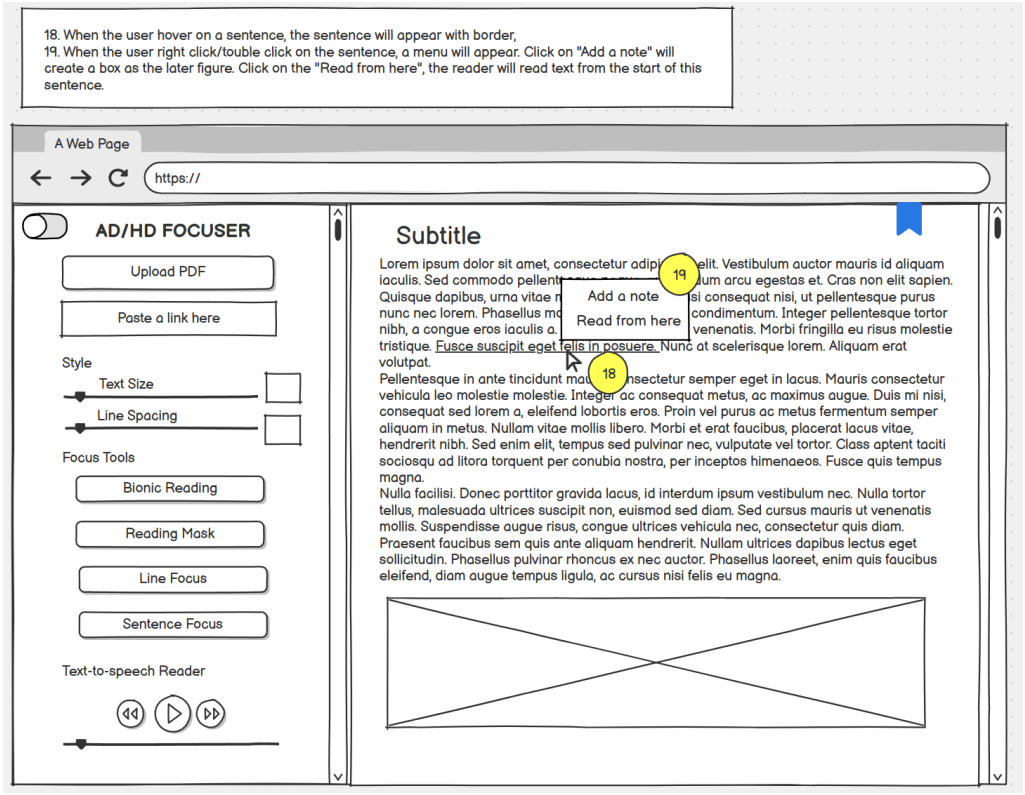
Bionic Reading (Welkin)



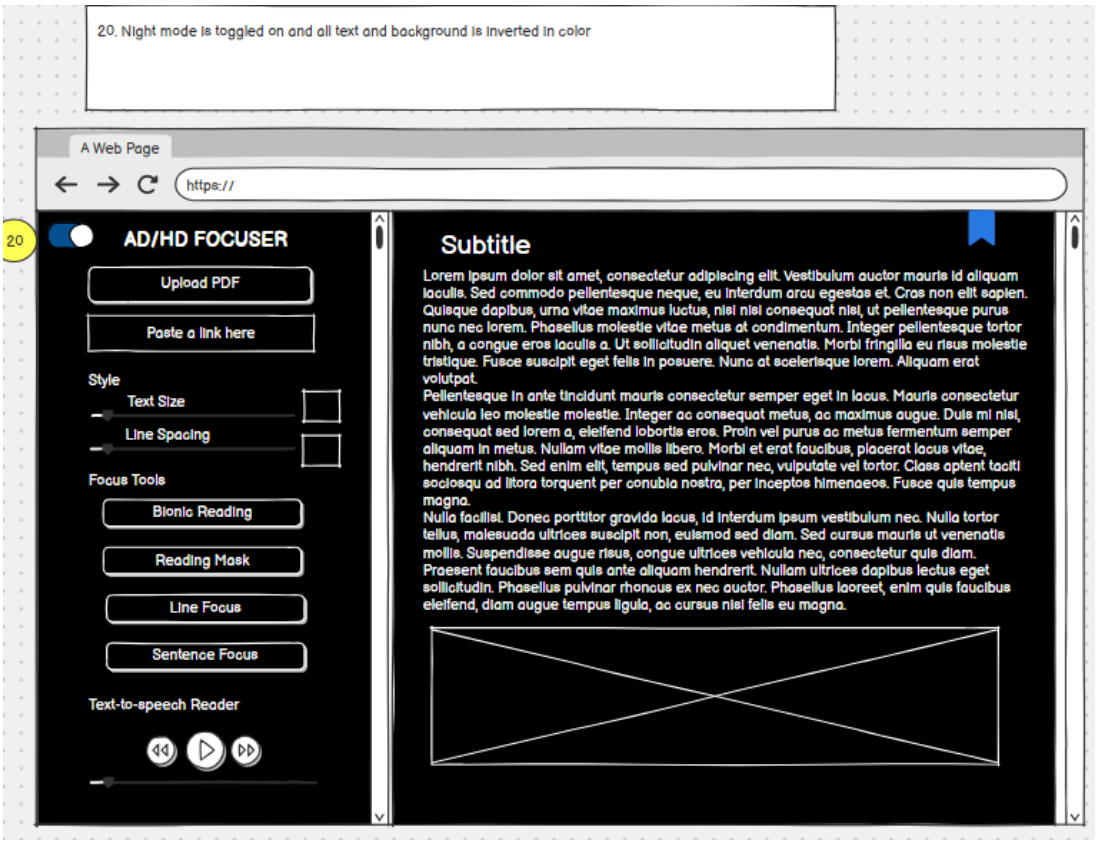
Reading Mask (Welkin)



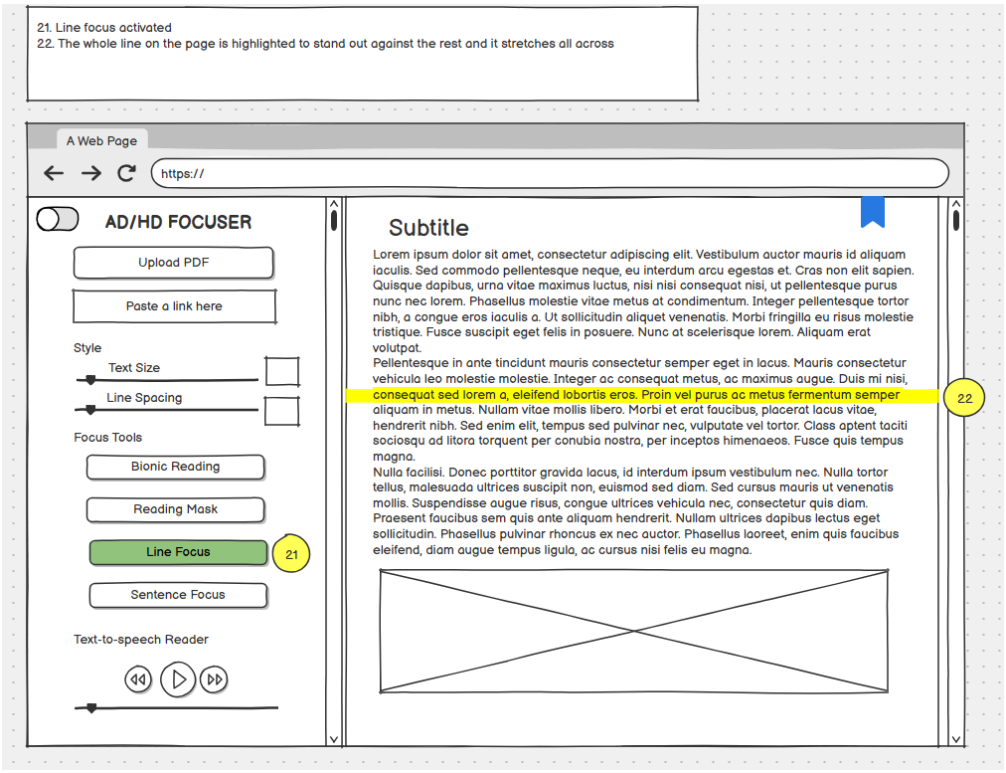
Sentence Hover (Welkin)



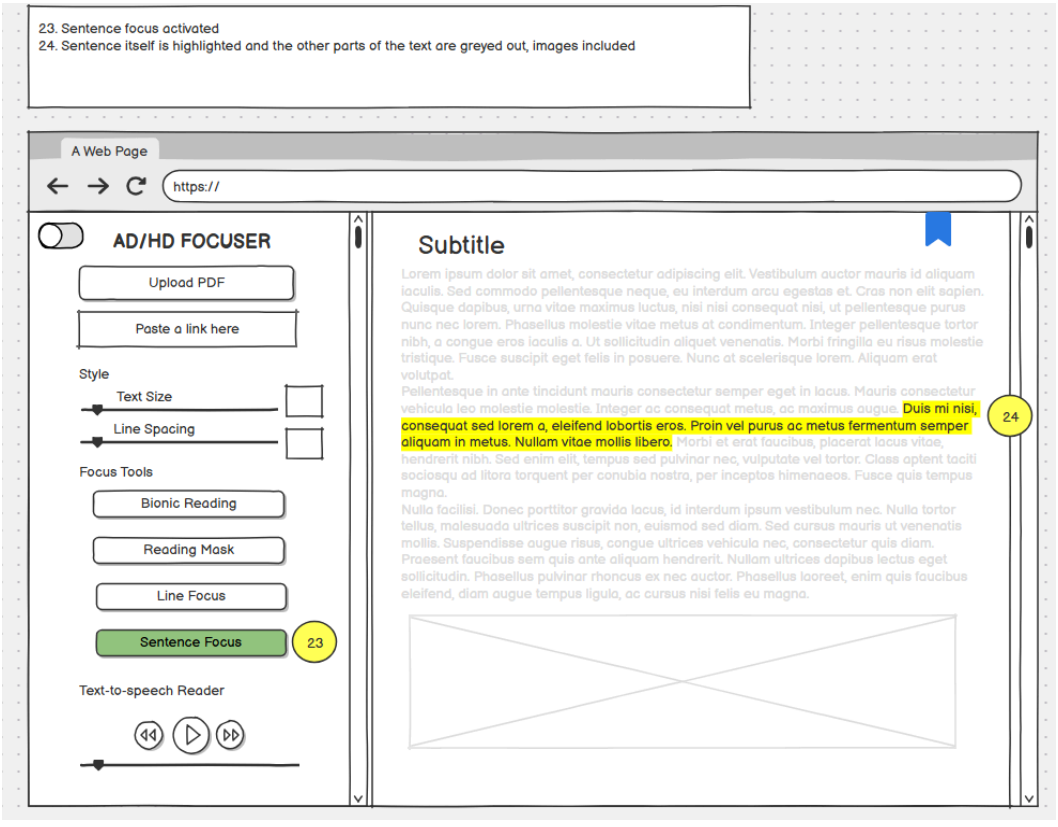
Night switch (Branden）



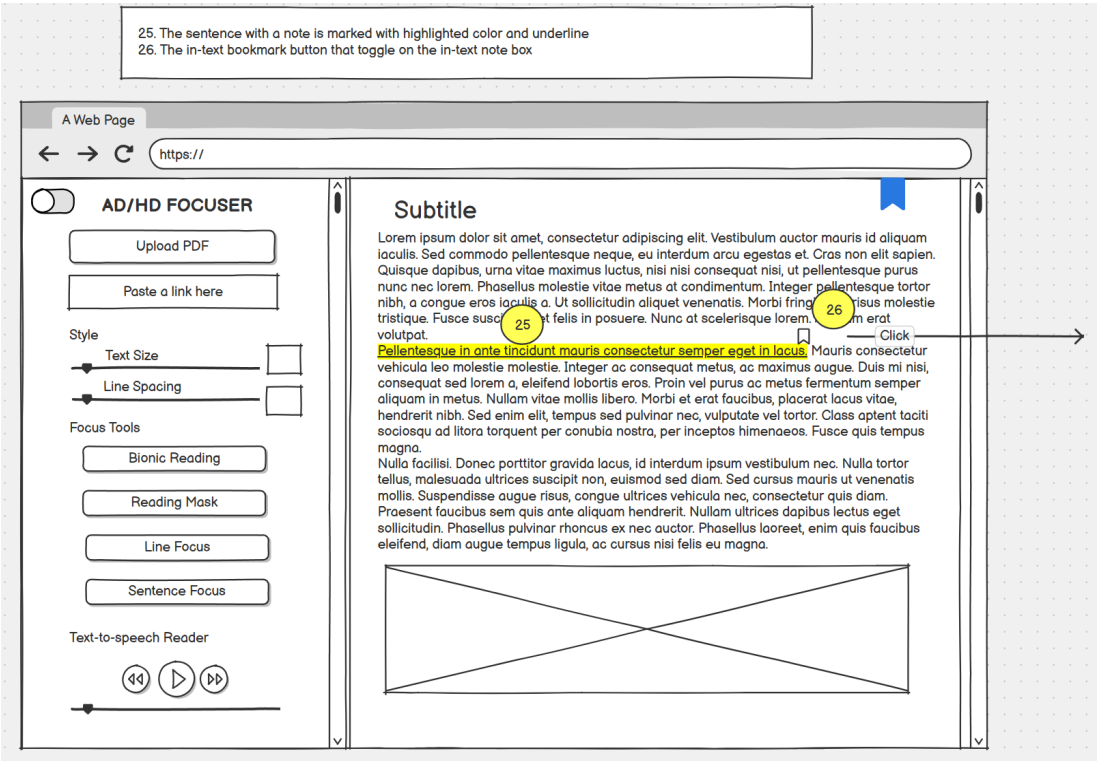
Line Focus (Branden)

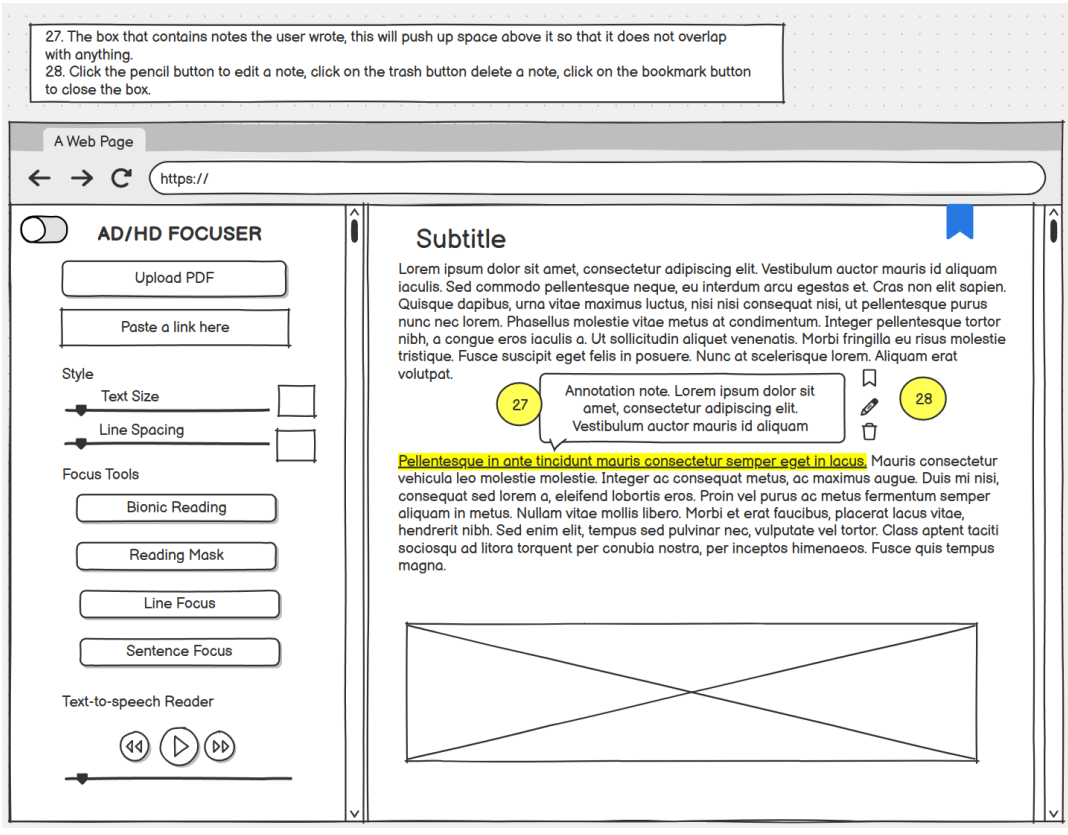


Sentence Focus (Branden)

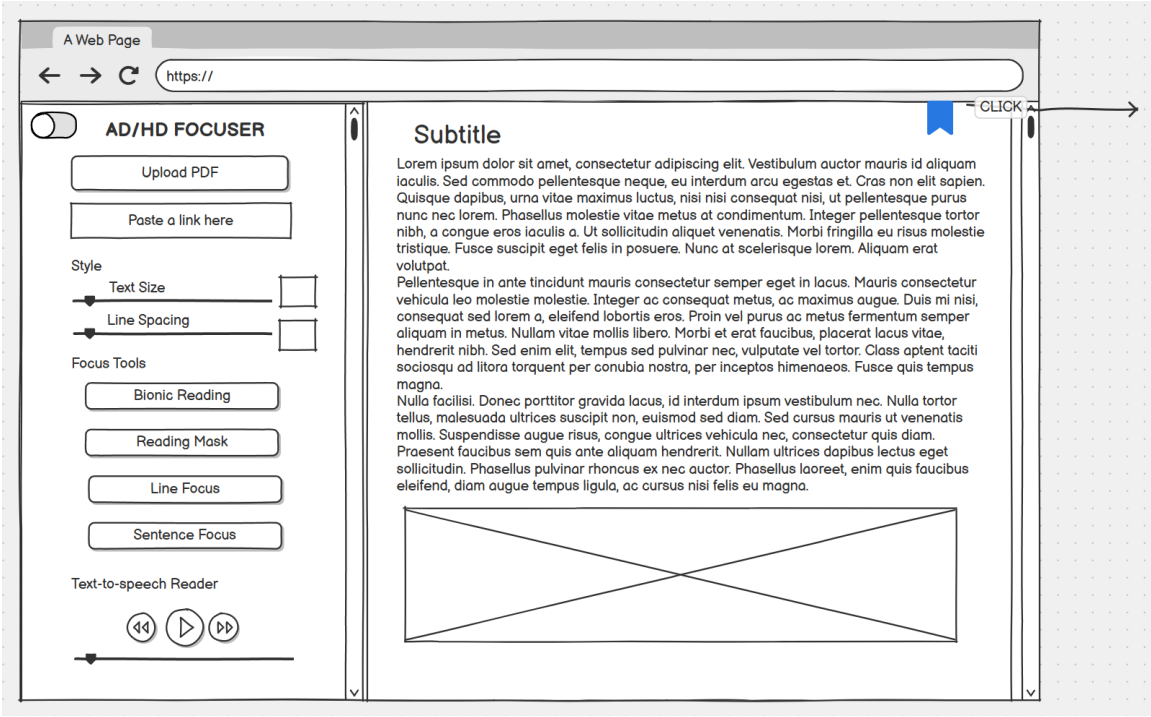


Bookmark(In text notes) (Welkin)

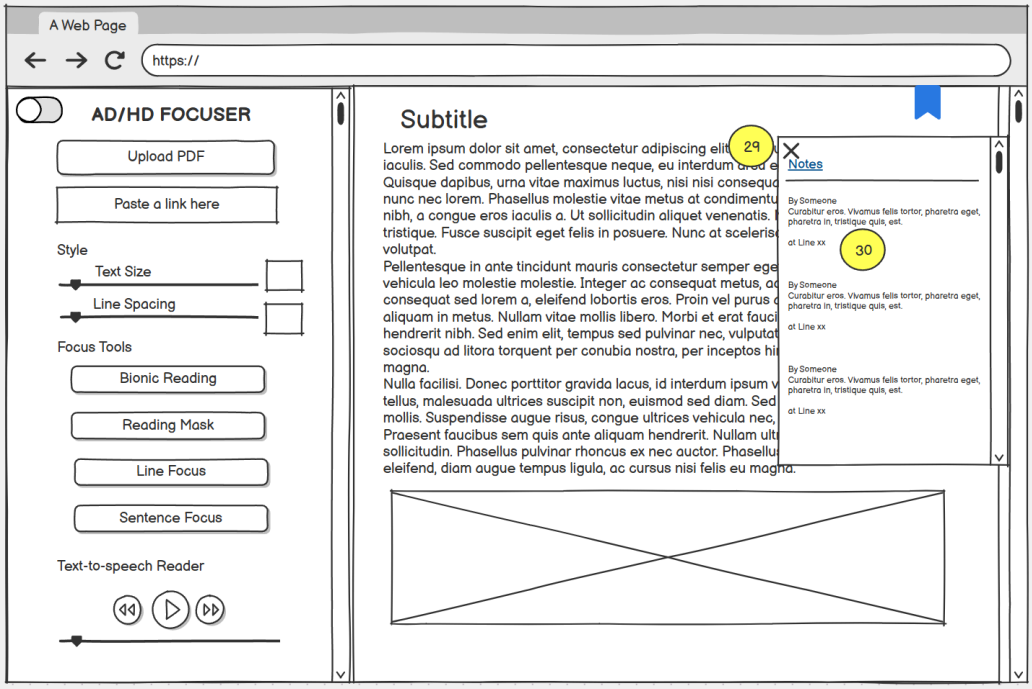




Bookmark(Notes outline) (Welkin)



29. Click on the blue bookmark button on the top right corner to open the Notes panel, with all the notes the user input listed
30. Each note consist of a skip link to skip to the line where the note is



Technology Specification

- **Programming Languages**
  - JavaScript
  - CSS
  - HTML
- **Libraries**
  - jQuery (JavaScript) – Used to find, select, and interact with webpage elements, including text highlighting and user interactions.
  - PDF.js (JavaScript) – Used to extract text, text positions, and render pages from PDF documents.
  - Mozilla Readability (JavaScript) – Used to extract the main content of webpages while removing unrelated elements such as advertisements and navigation menus.
  - DOMPurify (JavaScript) – Used to sanitize extracted HTML content before displaying it in the reader.
- **Frameworks**

  N/A
- **Existing Technologies**
  - OpenAI API – Used to analyze complex document layouts, identify reading order, and determine the relationships between text, figures, and captions.
  - Claude AI – The backup solution if we ended up unable to access the OpenAI API.

We will be creating a Google Extension which opens a website that allows users to work with large texts that may be dense in formatting. It has been an issue for the other existing ADHD text viewers such that, a web page ADHD reader cannot process text with images, and a PDF reader makes numerous mistakes when reading a scholarly article that contains graphical content and is formatted as two columns or more. Therefore, our plan of integrating chatGPT and other modern tools to help extract and sort the content from both webpages and PDFs will aim to provide the following functionality and more:

- Extract text from a PDF or a link that its output is accessible and contains rich text.
- Sort the text and any graphics that are considered the main content (contents that are not advertisements, or simply for decoration) in the correct order and display it to the user.
- Allow users to rescale the text, spacing, and theme, so that they can self-adjust the visual elements depending on their unique needs.
- Provide focus tools that operate correctly as the wireframe presented.

- Allow the user to annotate the text, and is able to save it, allowing them to navigate to their notes at any time.
- Allow the user to listen to the text with the ability to adjust the speed, control where to start reading, pausing, and navigate over text using the “go back one sentence” button and “go forward one sentence” button.

Design Specification

- ## Aesthetic Guidelines

  We plan to keep the layout of the extension clean and neat, minimizing the use of many colors because the reading masks are going to be the main component that should attract the user’s attention or distract them. We use underline and highlight colors with a bookmark symbol button to indicate the sentence contains a note. We also use green color to indicate what focus tool the user is using, and because the panel is mainly white and black, the color is also a unique shade which eliminates confusion for colorblind users. The semi-transparent mask, highlighted line, sentence and bionic reading are the main functional aesthetic components of the extension.

- ## Functional requirements

  We aim to provide a stable, consistent and reliable focuser for the user who suffers from reading difficulties. We do not aim to read or summarize text for the user, but cultivate an ADHD-friendly environment, such that the user will be able to complete reading on their own. We must allow the input file to be any formatted essay, single column, two-column, three-column, with charts, diagrams or images, and the text, images should present in the correct order as a single column. We plan to do so with the assistance of ChatGPT’s API, let it process the input file and identify which chunk of text will come first, and deal with text break-up when a page ends.

- ## UX Considerations

  We aim to make the experience as comfortable and easy to use for all users whether they use a screen reader or not. The addition of a night mode is to suit the eyes for those without vision issues as staring at a white screen for so long could cause eye strain. All buttons and such are conveniently located at the left for better reach without all sorts of buttons being all over the place on the screen. Buttons will also allow the user to know when they are being used with each button having its own distinct function such as line focus having the whole line highlighted across the screen and sentence focus being highlighted and having other sentences and surrounding objects be greyed out.

- ## Interaction Patterns

Users will be able to access all functions on the webpage through the use of a keyboard so that using a mouse is not the only way to toggle between certain functions on the left side of the page. Functions will be able to be turned on or off with the push of a button and when heading to the text or sentence being focused on, we plan to make it so that the keyboard user will be able to use the tab key directly to the area the user would like to focus on for convenience.

- **Feedback and Response Mechanism**

  With our usage of highlighted elements, it should be easy to show where the area is that a user would like to focus on. For those with a screen reader or prefer to use a screen reader when using the website, we plan to make the buttons say not only their purpose but also what they are specifically used for in case one is not familiar with what they are. Said buttons should be able to allow the user to focus on certain specified parts of the text they highlight such as a specific line or sentence. There is also a text reader that can also be toggled with a keyboard with the tab key in order to play or replay the section they wish to hear. There is a back and forward key for the user to replay certain parts as they wish and there is a speed slider to change the speed of the text being read.

Roles and Contributions

- **Welkin:**
  - **Contributions:**
    - Designed wireframes for default, bionic reading, reading mask, bookmark frames
    - Studied how existing technologies will help us in text extraction and conversion
    - Tested the existing ADHD readers and recorded their strengths and weaknesses
  - **Strengths:**
    - Knowledge in web design and user interface, front end development, has experience integrating AI with web pages, has experience writing skills for codex. Currently doing research on API usage and helping with the implementation.

- **Branden:**
  - **Contributions:**
    - Researched more on how the ChatGPT API works and how it could be utilized in the project
    - Contributed to the design of the Dark mode, line focus and sentence focus on their respective wireframes

        - Took note of previous ADHD readers and what we could innovate upon with ours with the modern technology we plan to utilize
    - **Strengths:**
        - Knowledge in using all 3 coding languages listed above (Javascript, HTML and CSS), Back end development, decent knowledge of ChatGPT's API, will help with the usage of the extensions with the help of available documentation of each on the internet